ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

POWER MACHINES N6

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NATED Engineering Studies, N6. Steam boilers, steam power plant (Rankine cycle) and turbines, internal combustion engines, air compressors and refrigeration. All property values needed are given in the questions; no steam tables are required.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations. Write units with all answers.
5. Enthalpy and other property values are given in the questions; steam tables are not needed.
6. Take the latent heat of steam at 100 °C as 2 257 kJ/kg.
7. Round off answers to TWO decimal places, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1: STEAM BOILERS

1.1 A boiler produces 5 000 kg of steam per hour with an enthalpy of 2 800 kJ/kg from feedwater with an enthalpy of 335 kJ/kg. It burns 650 kg of coal per hour with a calorific value of 28 000 kJ/kg. Calculate:

1.1.1 the heat transferred to the steam per hour (3)

1.1.2 the thermal efficiency of the boiler (3)

1.1.3 the equivalent evaporation "from and at 100 °C" per hour (3)

1.1.4 the equivalent evaporation per kilogram of coal (3)

1.2 State the function of each of the following boiler accessories: (a) economiser (b) superheater (c) air preheater (6)

1.3 Name TWO boiler mountings that are required for safety. (2)

[20]

QUESTION 2: STEAM PLANT AND TURBINES

2.1 In a steam power plant working on the Rankine cycle, steam enters the turbine with an enthalpy of 3 230 kJ/kg and leaves with an enthalpy of 2 210 kJ/kg. The condensate leaves the condenser with an enthalpy of 192 kJ/kg, and the feed pump work is 4 kJ/kg.

2.1.1 Draw a labelled block diagram of the plant (boiler, turbine, condenser and feed pump). (3)

2.1.2 Calculate the turbine work and the net work per kilogram of steam. (3)

2.1.3 Calculate the heat supplied in the boiler per kilogram of steam. (2)

2.1.4 Calculate the thermal efficiency of the cycle. (2)

2.1.5 Calculate the steam flow rate needed for a net power output of 30 MW, and the specific steam consumption in kg/kWh. (4)

2.2 Explain the difference between an impulse turbine and a reaction turbine. (4)

2.3 Why are large steam turbines compounded? Name ONE method of compounding. (2)

[20]

QUESTION 3: INTERNAL COMBUSTION ENGINES

3.1 A four-cylinder, four-stroke petrol engine has a bore of 100 mm and a stroke of 120 mm. At 3 000 r/min, the indicated mean effective pressure is 750 kPa and the brake torque is 180 N·m. The engine uses 16 kg of fuel per hour with a calorific value of 44 000 kJ/kg. Calculate:

3.1.1 the indicated power (5)

3.1.2 the brake power (3)

3.1.3 the mechanical efficiency (2)

3.1.4 the brake thermal efficiency (4)

3.1.5 the brake specific fuel consumption in kg/kWh (2)

3.2 Give TWO differences between a petrol (spark-ignition) engine and a diesel (compression-ignition) engine. (4)

[20]

QUESTION 4: AIR COMPRESSORS

4.1 A single-stage, reciprocating air compressor takes in 2 m³ of free air per minute at 100 kPa and 20 °C and compresses it to 800 kPa according to the law $pV^{1,3} = C$. Calculate:

4.1.1 the indicated power required (5)

4.1.2 the isothermal power (3)

4.1.3 the isothermal efficiency (2)

4.1.4 the delivery temperature of the air in °C (4)

4.2 Explain THREE advantages of multi-stage compression with intercooling. (6)

[20]

QUESTION 5: REFRIGERATION

5.1 A vapour-compression refrigeration plant has the following enthalpies: refrigerant leaving the evaporator 395 kJ/kg; leaving the compressor 425 kJ/kg; leaving the condenser 250 kJ/kg. The expansion through the throttle valve is at constant enthalpy. The plant must provide 10 kW of cooling.

5.1.1 Calculate the refrigerating effect per kilogram. (2)

5.1.2 Calculate the compressor work per kilogram and the coefficient of performance. (3)

5.1.3 Calculate the mass flow rate of the refrigerant. (2)

5.1.4 Calculate the compressor power and the heat rejected in the condenser. (4)

5.2 Draw a labelled block diagram of a vapour-compression refrigeration system and state the function of each of the FOUR main components. (9)

[20]

TOTAL: 100

END OF QUESTION PAPER